WINDOWS SERVER 2022

FILE SHARING & ACCESS CONTROL

Personal IT Project

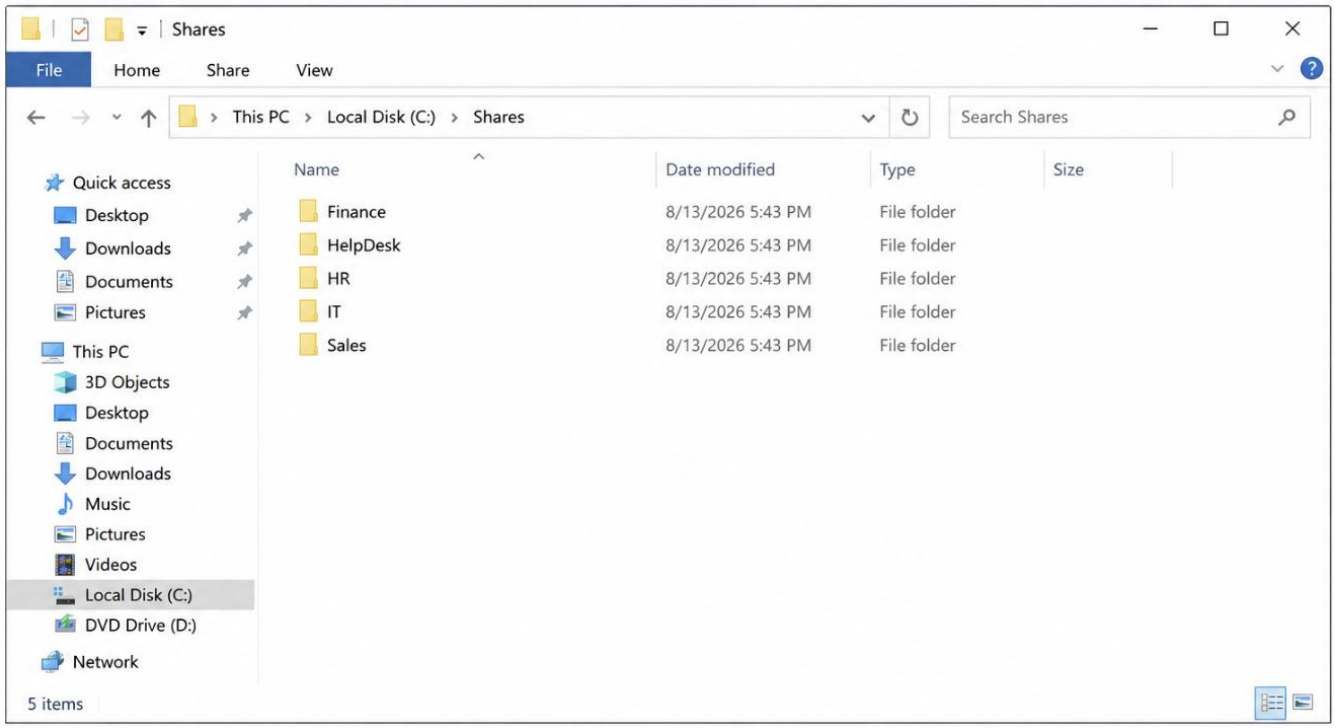
Project Focus

I expanded my existing Windows Server 2022 and Active Directory homelab by adding department file shares and access control. I configured the shares on DC01, used Active Directory security groups for access, and checked the results from a Windows 11 client.

Environment

Windows Server 2022 ● Windows 11 Client ● Active Directory ● VMware

Domain	Domain Controller	File Share Location
homelab. local	DC01	C:\Shares



DC01 showing the department share structure

1.Project Overview

This project adds Windows file services to the Active Directory lab I have already built. I created department folders, made them available as network shares, connected to security groups, and then tested the results from Windows 11.

Component	What I Configured
Existing Platform	Used the existing Windows Server 2022 domain controller, DC01, in the homelab.local domain.
File Share Location	Created the department folder structure C:\Shares
Department Shares	Created shares for Finance, HelpDesk, HR, IT, and Sales.
Access Control	Used Active Directory security groups with NTFS permissions to control folder access.
Client Testing	Used Windows 11 to open the shares, create test files, and check a denied access result.

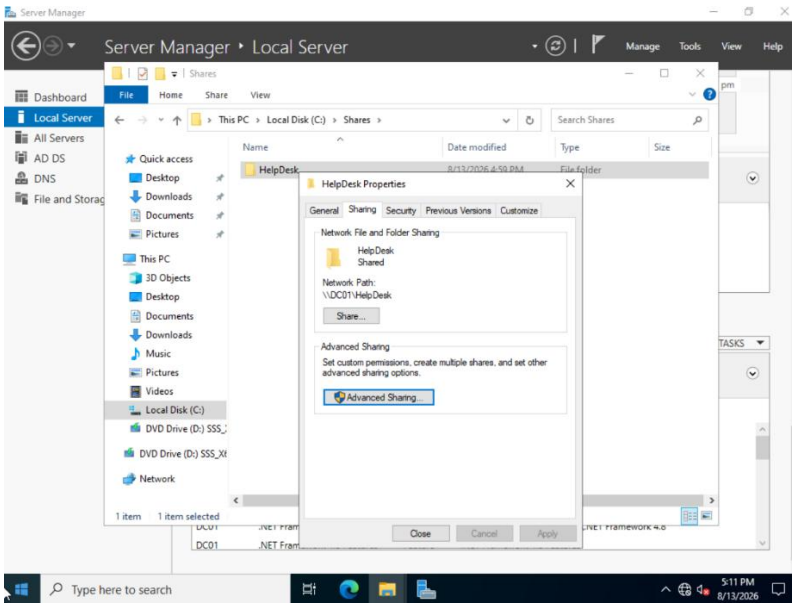
2.Access Control Design

I used a group based permission setup instead of assigning access directly to individual users. The domain account makes the request, the security group represents the department or role, and the folder's NTFS permissions control access.

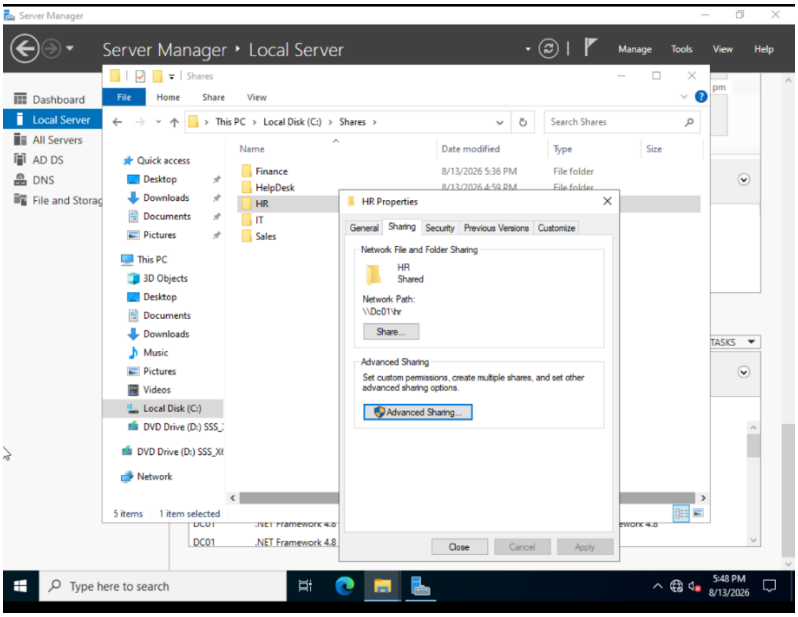
Layer	How I Used it
User Identity	The Windows domain account is the account making the access request.
Security Group	Department or role based groups are used so access can be managed through group membership.
Network Share	Each department folder is available through a Windows UNC path.
NTFS Permissions	The department security group is listed in the folder’s security settings
Client Validation	Windows 11 is used to confirm whether the share opens and whether a test file can be created

3. Department File Shares

I created the department folders under C:\Shares and configured them as Windows network shares. This allowed the folders to be accessed from the Windows 11 client using UNC paths.



HelpDesk share configuration on DC01



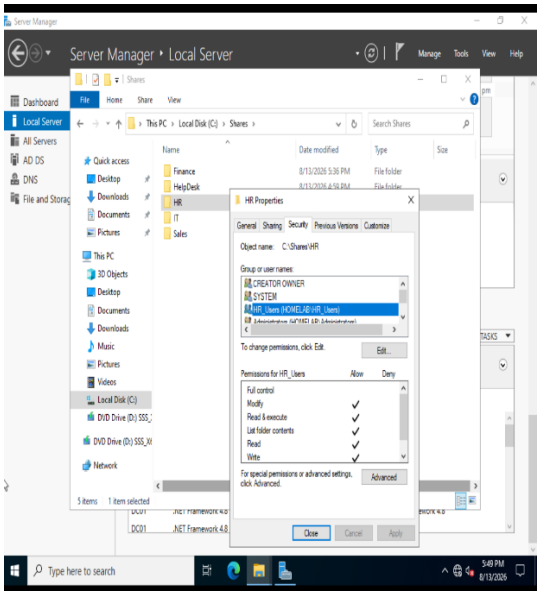
HR share configuration on DC01

Department	UNC Path
Finance	\\DC01\Finance
HelpDesk	\\DC01\HelpDesk
HR	\\DC01\HR
IT	\\DC01\IT
Sales	\\DC01\Sales

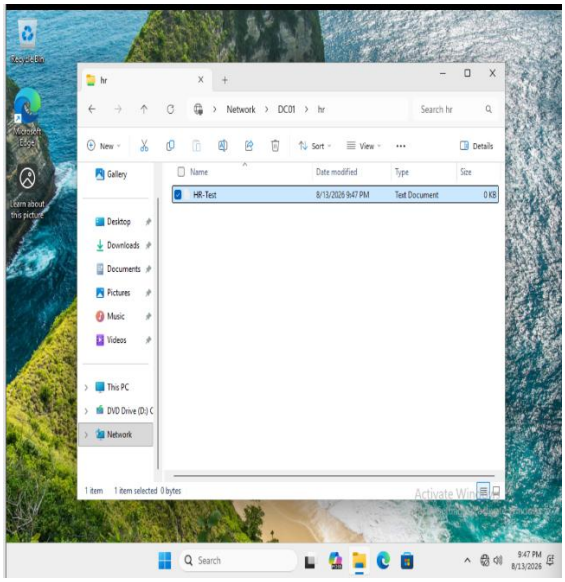
What this demonstrates: The department folders were configured as network shares and made available to the Windows 11 client through their UNC paths.

4. HR Access — Permissions and Validation

For HR, I used the HR_Users security group in the folder's NTFS Security settings. I then accessed the HR share from the Windows 11 client and created a test file. This provided both server side evidence of the permission configuration and client side proof that the access worked.



HR NTFS Security settings showing HR_Users.



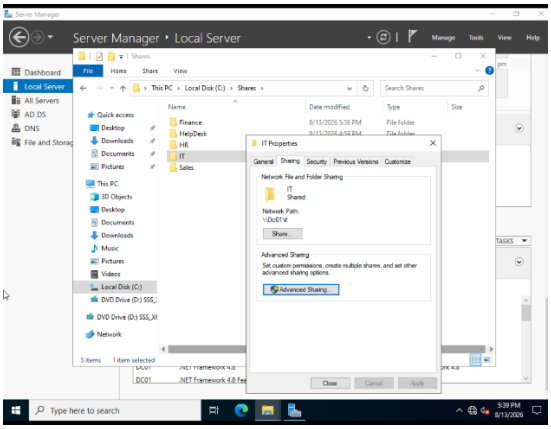
Windows 11 client accessing the HR share

Validation result: The HR share opened successfully from the Windows 11 client, and the HR Test text document was created.

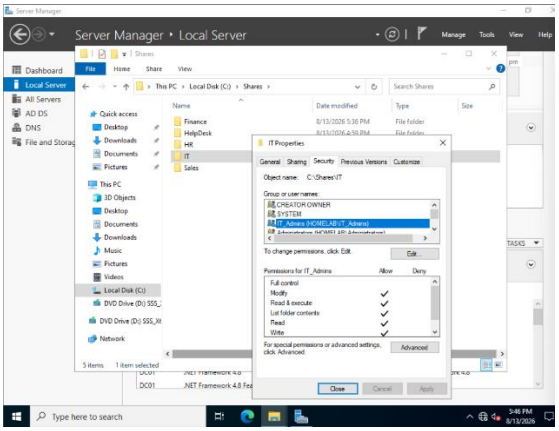
This test connects the NTFS permission configuration on the server to an actual access result from the client.

5. IT Access — Configuration and Client Test

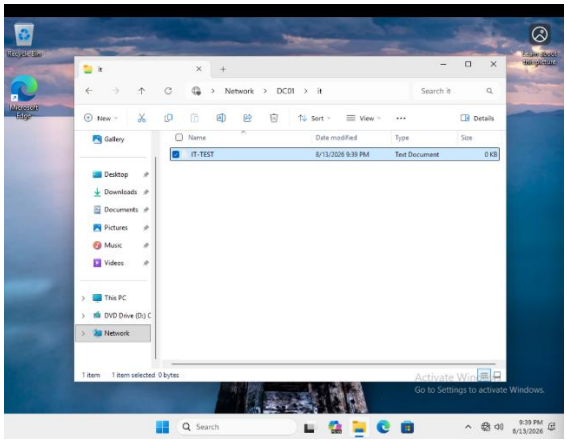
For the IT folder, I used the IT_Admins security group to control access. I configured the share, reviewed the NTFS Security settings, and then tested the share from the Windows 11 client by creating the IT-TEST file.



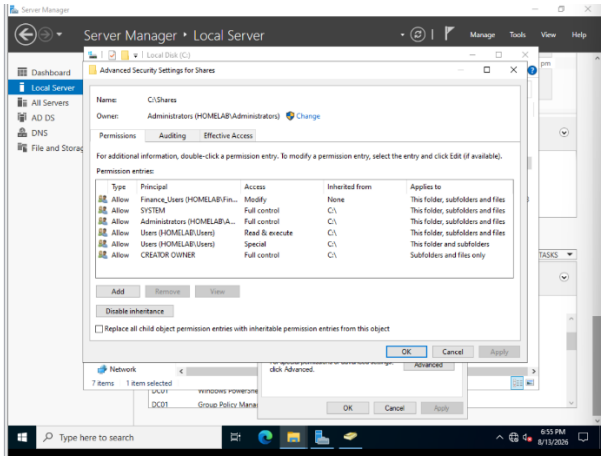
IT share configuration on DC01



IT NTFS security setting showing IT_Admins



Windows 11 Client creating IT-Test

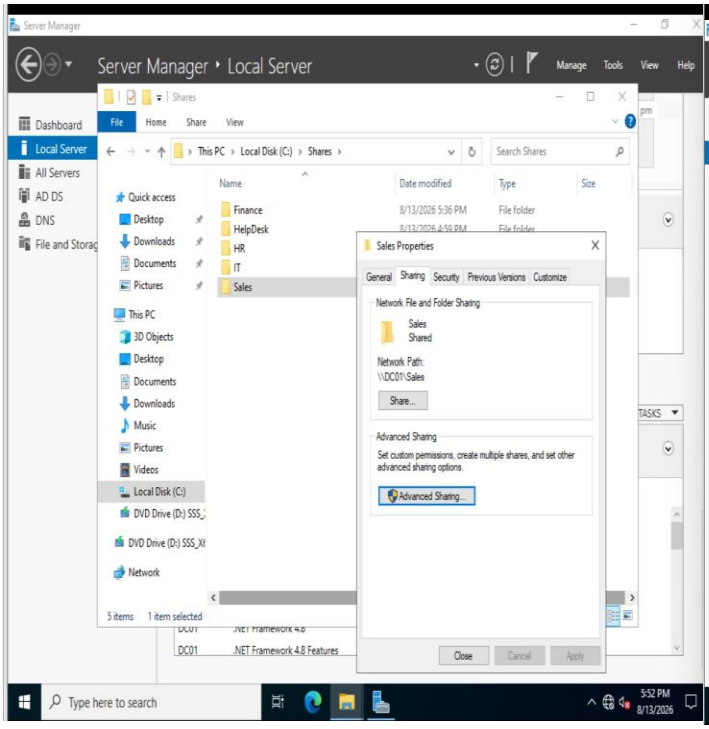


Advanced Security Settings for the shared folders

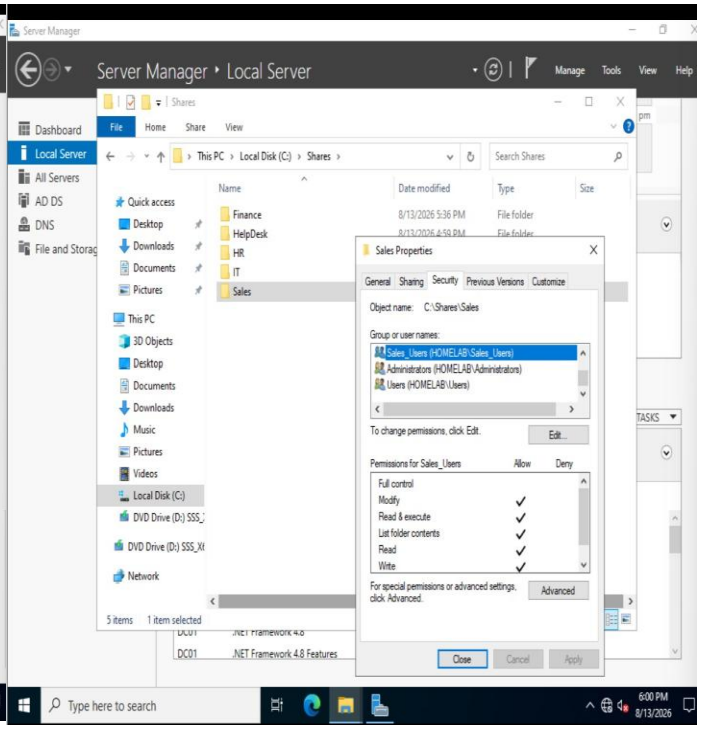
Validation Result: The IT share opened successfully from Windows 11, and the IT-TEST file was created. The server-side security settings provide supporting evidence of the configured permissions.

6. Sales Access — Permissions and Validation

For Sales, I used the Sales_Users security group to control folder access and configured the folder as a network share. I then tested the share from the Windows 11 client by creating a test file.



Sales share configuration on DC01



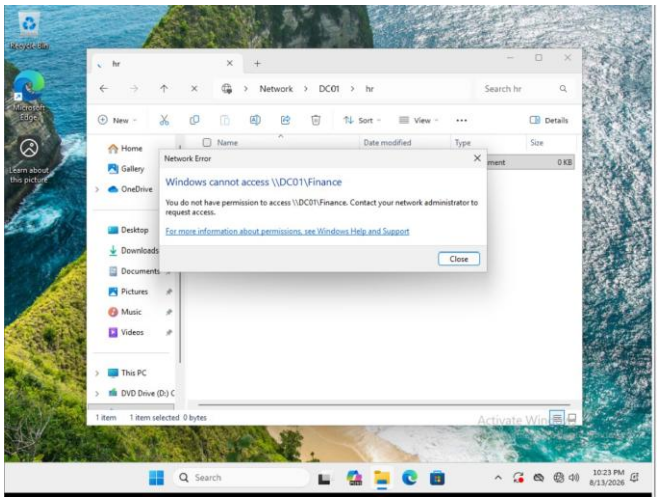
Sales NTFS Security Settings showing Sales_user

Validation Result: The Sales share opened successfully from the Windows 11 client, and the Sales-Test file was created successfully.

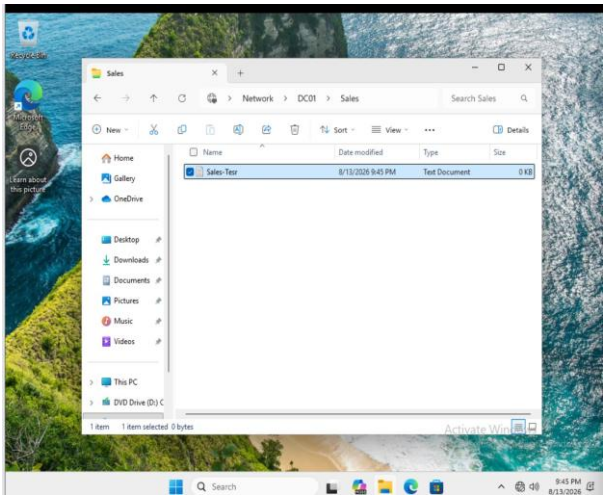
The server side settings show the configured access, while the Windows 11 client confirms that the share could be accessed and used.

7. Access Testing — Allowed and Denied Results

I tested the department shares from the Windows 11 client to verify that the permissions were working as expected. HR, IT, and Sales were able to access their shares and create test files, while the Finance test returned an access denied message.



Finance access denied



Sale share accessed and test file created

Technical Skills

Windows Server 2022	Configuring department folders and network shares on DC01
Active Directory	Working with users, security groups, and the existing domain
Security Groups	Using department groups to manage folder access
NTFS Permissions	Configuring folder level permissions for department access
Windows File Sharing	Creating and accessing department network shares
Access Control	Allowing department access and blocking unauthorized access
Windows 11 Administration	Testing the shares from the client workstation
Permission Testing	Verifying both successful and denied access